

Mangalore University

Syllabus of

STATISTICS

**A Major Subject for B.Sc. Degree Programme
(CBCS)**

(With effect from 2024-25 onwards)

Mangalore University

Revised syllabus of STATISTICS, a major subject for B.Sc. Degree Program

PREAMBLE

Mangalore University has adapted NEP in 2021-22 and it has been three years. Now as per the direction of the Government the revision of syllabus of Statistics, a major subject for the three years B.Sc. degree program is undertaken for the students studying 2024-25 onwards. Statistics is vigorously gaining importance and recognition in today's society. For a long time, Statistics was identified solely with the displays of data and charts pertaining to the economic, demographic, and political situations prevailing in a country. However, gigantic advances during the twentieth century have enabled statistics to grow and assume its present importance as a discipline of data-based reasoning. Statistics, as a scientific discipline, stretch well beyond the confines of data display. It deals with collecting informative data, analysis interpreting these data, and drawing conclusions about a phenomenon under study. The scope of this subject naturally extends to all processes of acquiring knowledge that involve fact finding through collection and examination of data. These days Data Science emerges as a prominent and much used subject whose foundation is Statistics and it is advanced with algorithmic approach. In this programme, a foundation is given. Also, Statistics plays vital role in Big-data and Analytics, which is now an emerging field in all spheres of human activity. There is a great demand for Data Science and data analytics.

Accordingly as per CBCS Semester scheme the board designed the curriculum of B.Sc. Statistics keeping in view that the curriculum of B.Sc. Statistics should be flexible enough to prepare graduates to either be functioning Statisticians or to go on to post-graduation studies

The following changes were made in the curriculum.

- 1) The Programme of study consists of Core Courses and Elective courses.
- 2) Core courses will consist of compulsory/major courses which are compulsorily studied by a student in all the six semesters. Elective courses are opted in I to IV Semesters.
- 3) Statistical computing is taught using R software.
- 4) As any career in Statistics after B. Sc. requires working with computer systems, students learn to solve the problems using computers. Introduction of computer based computation using EXCEL and R programming.
- 5) Description of various multivariate data analysis methods will helps them in data analysis using computers. This will help them in career and any data analysis.

ELIGIBILITY

1. Only those candidates who have passed Pre-University Course in Science or an equivalent course with Mathematics as one of the subject are eligible to take Statistics as one of the major subjects in B.Sc. Course.
2. Any student taking Statistics as one of the major subjects in the B.Sc. course shall take Mathematics as another major subject.

Programme Outcomes (POs)

By the end of the program the students will be able to:

1. Acquire fundamental/systematic or coherent understanding of the academic field of Statistics and its different learning areas and applications.
2. Develop and demonstrate an ability to understand major concepts in various disciplines of Statistics.
3. Demonstrate the ability to use skills in Statistics and different practicing areas for formulating and tackling Statistics related problems and identifying and applying appropriate principles and methodologies to solve a wide range of problems associated with Statistics.

4. Understand procedural knowledge that creates different types of professionals related to subject area of Statistics, including professionals engaged in government/public service and private sectors.
5. Plan and execute Statistical experiments or investigations, analyze and interpret data/information collected using appropriate methods, including the use of appropriate statistical software including programming languages, and report accurately the findings of the experiment/investigations.
6. Have a knowledge regarding use of data analytics tools like Excel and R-programming.
7. Developed ability to critically assess a standard report having graphics, probability statements.

Course Articulation Matrix: Mapping of Course Outcomes(COs)with Program Outcomes (POs 1-7)

Course Outcomes (COs)/ Program Outcomes (POs)	1	2	3	4	5	6	7
CO-1. Acquire knowledge of introductory statistics, its scope and importance in various are as such as Medical, Engineering, Agricultural and Social Sciences etc.	x			x	x		
CO-2 2.Will learn various types of data, their organization and descriptive statistics such as presentations in tabular form graphs and summary measures such as measures of central tendency and dispersion etc.		x	x	x	x		

Scheme of Instruction/Examination:

The board recommends that the existing scheme of instruction and examination in theory and practical may be continued. The theory question paper in all semesters shall cover all the topics in the pertaining syllabus with proportional weightage to the number of hours of Instruction prescribed.

2 questions from each unit with internal choice with or without sub-questions.

Practical Examination:

- 1) Students will have to answer any four questions out of 8 questions. All questions carry equal marks.
- 2) Practical paper in each semester carries 50 marks and the split up of the practical marks are as given below.
 - a) Three hour examination with two examiners which includes at least one external examiner carries 30 marks.

b) Ten marks for class records. c) Five marks each for the preparatory practical examination using calculator and for Statistical computing using R software. In the semesters where R language is not discussed, ten marks to be allotted for the preparatory practical examination.

Semester I								
Sl. No	Course Code	Title of the Course	Category of Courses	Teaching Hours per Week	SEE	IA	Total Marks	Credits
1		Descriptive Statistics	Theory	4	80	20	100	3
2		Statistics Practical - I	Practical	4	40	10	50	2
Semester II								
3		Probability Theory and Probability Distributions	Theory	4	80	20	100	3
4		Statistics Practical-II	Practical	4	40	10	50	2

DESCRIPTIVE STATISTICS

Unit–1: Introduction to Statistics

10 Hours

Statistics: Definition and scope. Data: quantitative and qualitative, cross-sectional and time series, discrete and continuous. Scales of measurement: nominal, ordinal, interval and ratio. Presentation of data: tabular and graphical. Frequency distributions, cumulative frequency distributions and their graphical representations. Stem and leaf displays. Concepts of population and sample. Different methods of data collection:- SRS, Stratified, Systematic and Cluster sampling methods: definitions only.

Unit–2:Univariate Data Analysis

18 Hours

Concept of measures of central tendency: Mean, weighted mean, trimmed mean, Median, Mode, Geometric and harmonic means, properties, merits and limitations, relation between these measures. Partition Values: Quartiles, Deciles and Percentiles. Measures of Dispersion: Range, Quartile deviation, Mean deviation, Variance, Standard deviation and their relative measures. Moments, Skewness and Kurtosis. Outliers, Box Plot.

Unit –3: Bivariate Data Analysis

18 Hours

Bivariate Data, Correlation: Scatter plot, Karl Pearson's correlation coefficient, Spearman's rank correlation coefficient, properties and results. Curve fitting, concept of errors, principle of least squares, fitting of polynomial and exponential curves. Analysis of Categorical Data: Contingency table, independence and association of attributes, measures of association- odds ratio, Pearson's and Yule's measure.

Unit -4: Statistical computing (R Software)

10 Hours

Introduction to R, R as a calculator, statistical software and a programming language, R preliminaries, getting help, data inputting methods(direct and importing from other spread sheet applications like Excel), data accessing, and indexing, packages, Graphics in R, built in functions, saving, storing and retrieving work. Descriptive statistics: diagrammatic representation of univariate and bivariate data (box plots, stem and leaf diagrams, bar plots, pie diagram), measures of central tendency (mean, median and mode), partition values, measures of dispersion (range, standard deviation, mean deviation and inter quartile range), summaries of a numerical data, skewness and kurtosis.

References:

1. Agresti, A.(2010).Analysis of Ordinal Categorical Data,2nd Edition,Wiley.
2. Anderson T.W. and Jeremy D. Finn (1996).The New Statistical Analysis of Data, Springer
3. Gupta, S. C. (2018). Fundamentals of Statistics, Himalaya Publishing House, 7th Edition.
4. Gupta S.C. and V.K. Kapoor (2020). Fundamentals of Mathematical Statistics, Sultan Chand and Co. 12th Edition.
5. Hogg, R. V. McKean J. W. and Craig, A. T. (2012). Introduction to Mathematical Statistics, Pearson 7th Edition.
6. Johnson, R.A. and Bhattacharyya, G.K. (2006). Statistics: Principles and methods. 5th Edition, John Wiley & Sons, New York.
7. Medhi, J.(2005).Statistical Methods, New Age International.
8. Ross, S.M. (2014). Introduction to Probability and Statistics for Engineers and Scientists, 5th Edition, Academic Press.
9. Tukey, J.W.(1977). Exploratory Data Analysis, Addison-Wesley Publishing Co.
10. Sudha G. Purohit, Sharad D. Gore, Shailaja R Deshmukh,(2009). Statistics Using R, Narosa Publishing House.
11. Emmanuel Paradis (2005). R for Beginners (available at https://cran.rproject.org/doc/contrib/Paradisrdebuts_en.pdf)

Practical Content based on Descriptive Statistics:

(Carrying-out the practical using calculator)

1. Presentation of data by frequency tables, diagrams and graphs, stem and leaf chart.
2. Arithmetic Mean (AM), geometric mean, harmonic mean, weighted AM, trimmed mean, corrected mean.
3. Median, mode and partition values.
4. Quartile deviation, mean deviation- absolute measures and relative measures, Box plot.
5. Standard deviation, coefficient of variation and, combined variance
6. Problems on moments, skewness and kurtosis.
7. Fitting of curves by least squares method.
8. Product moment correlation coefficient and rank correlation.
9. Problems on Association of attributes.
10. Statistical Computing using R software- Descriptive Statistics.

PROBABILITY THEORY AND PROBABILITY DISTRIBUTIONS

UNIT I - Probability

10 Hours

Random Experiments, Sample space, Elementary events and compound events, Algebra of events, Classical definition of probability and its limitations, relative frequency approach. Axioms of probability, Deduction of classical definition from Axiomatic definition and other results. Addition theorem, Conditional Probability and Independence, Multiplication theorem, Bayes' Theorem (with proof) and its applications.

UNIT II - Random Variables

15 Hours

Random variables (discrete and continuous), properties, probability mass function, probability density function their properties, probability distribution function - properties. Joint density functions, marginal density functions and conditional density functions. Expectation of random variables - rules of expectation, addition and multiplication theorems of expectation, variance, covariance and correlation. Mean and covariance of linear combination of random variables, moments, measures of location and dispersion - skewness and kurtosis of a random variable, moment generating function (MGF), and its properties, cumulant generating function (CGF), cumulants. Chebyshev's inequality.

UNIT III – Standard Discrete Probability distributions

15 Hours

Bernoulli, Binomial, Poisson, discrete uniform, Geometric, Negative Binomial, Hyper geometric distributions- definition through probability mass function (p.d.f.) - Mean, Variance, MGF, CGF, Recurrence relation between moments, properties, inter relationships.

UNIT IV – Standard Continuous Probability distributions

16 Hours

Uniform, exponential, gamma (one parameter and two parameters), beta (first kind and second kind), Cauchy, normal- definition through p.d.f., mean, variance, MGF, CGF, moments and properties.

References:

1. Dudewitz. E. J. and Mishra.S.N.(1998).Modern Mathematical Statistics. John Wiley.
2. Goon A.M., Gupta M. K. ,Das Gupta.B. (1991), Fundamentals of Statistics, Vol. I, World Press, Calcutta.
3. Hogg R, V., Mckean J.W, and Craig, A.T (2019). Introduction to mathematical Statistics, 8th Edition, Pearson Education, New Delhi.
4. Hogg, R.V., Tanis, E.A. and Rao J.M.(2009). Probability and Statistical Inference, Seventh Edition, Pearson Education, New Delhi.
5. Mood, A.M., Graybill, F.A. and Boes, D.C. (2007). Introduction to the Theory of Statistics, 3rd Edition. (Reprint), Tata McGraw-Hill Pub.Co. Ltd.
6. Ross, S. (2002), A First Course in Probability, Prentice Hall.

List of Practical Assignments

Practical Content based on Probability Theory:

(Carrying-out the practical using calculators and R Software)

- 1) Probability- Addition theorem, multiplication theorem and conditional probability.
- 2) Probability- Bayes' theorem.
- 3) Random variables (Univariate), pmf, pdf and Distribution functions, mean and variance.
- 4) Bivariate Probability Distributions - Conditional Mean, Conditional Variance, Correlation and Chebyshev's inequality.
- 5) Exercise on Binomial distribution.
- 6) Exercise on Poisson distribution.
- 7) Exercise on Geometric, Negative Binomial and, Hyper geometric distribution.
- 8) Exercise on normal distribution.
- 9) Exercise on exponential distribution and Weibull distribution.

Outline of Syllabus for III, IV, V and VI Semesters.

III Sem: Sampling – includes simple random sampling, Stratified, Systematic, Transformation of random Variables, Sampling distributions, that is chi, t F

IV Sem: Statistical Inference I – includes Point Estimation, Testing of Hypothesis, Interval estimation, Standard tests (like t, F, chi square), Likelihood ratio tests

Elective Paper will include topics – Official Statistics, Index numbers , Time Series, Demography, Vital Statistics

V Sem : 1) Includes Linear Regression, Logistic regression, Factor Analysis, Cluster, Discriminant, Canonical, SEM
2) Analysis of Experimental Designs CRD, RBD, LSD, YSD, Factorial, Missing plot Technique, Some part of Applied Statistics

VI Sem: Details can be given later 1) SQC and LPP 2) Introduction Data Science and Data Analytics